

3 Method of Types

Reading: Chapter 11.1–11.5 of Cover and Thomas

Review

- Weak AEP (WLLN applied to $-\log P(X)$)
- Weak typical set: equiprobable sequences and the size of the weak typical set.
- Exponential approximations
- Lossless source coding: achievability and converse

Preview of Lec. 10–12

- Method of types
- Strong typicality and strong AEP
- Universal coding

3.1 Types

In the last two lectures, we talked about weak typical set, which is defined as $A_\delta^{(n)}(P) = \{x^n \in \mathcal{X}^n : e^{-n(H(P)+\delta)} \leq P^n(x^n) \leq e^{-n(H(P)-\delta)}\}$. Consider the following two examples of distributions: e.g. Bern(1/3) and Bern(1/2). For Bern(1/3), the typical set is much smaller than the entire set $\{0, 1\}^n$. But for Bern(1/2), since every binary sequence of length n occurs with the same probability 2^{-n} , the typical set is equal to the entire set $\{0, 1\}^n$. But intuitively, when we call a sequence typical, we expect the empirical distribution of the sequence matches with the probabilistic distribution. Therefore, Bern(1/2) example shows some weakness of the definition of the weak typical set. In this lecture, we will introduce another way to define typicality based on empirical distribution of a sequence.

Definition 1 (Empirical distribution). For $x^n \in \mathcal{X}^n$, we write $\hat{P}_{x^n}$ as the empirical distribution of x^n .

$$\begin{aligned}\hat{P}_{x^n}(a) &= \frac{1}{n} \sum_{i=1}^n \mathbb{1}(x_i = a), \quad \forall a \in \mathcal{X} \\ &= \frac{1}{n} N(a|x^n)\end{aligned}\tag{1}$$

where $N(a|x^n)$ is the number of a 's in the given string x^n .

Definition 2. Let $\mathcal{P}_n$ be the set of all valid empirical distributions over $\mathcal{X}$ for a sequence of length n .

Example: If $\mathcal{X} = \{0, 1\}$,

$$\mathcal{P}_n = \left\{ [1, 0], \left[\frac{n-1}{n}, \frac{1}{n} \right], \left[\frac{n-2}{n}, \frac{2}{n} \right], \dots, [0, 1] \right\}.\tag{2}$$

Definition 3 (Type). For a distribution $P \in \mathcal{P}_n$, the type- P is a subset of $\mathcal{X}^n$, containing all length- n sequences of empirical distribution P :

$$T_P \triangleq \{x^n \in \mathcal{X}^n : \hat{P}_{x^n} = P\}.\tag{3}$$

Example:

$$\left| T_{\left[\frac{k}{n}, \frac{n-k}{n}\right]} \right| = \binom{n}{k}. \quad (4)$$

The space of $\mathcal{X}^n$ can be divided into types, each containing all the sequences with a certain empirical distribution. Some types are (exponentially) larger than the others. Our plan is to compute how large different types are and the probability that a certain sequence of DMS(Q) falls in a given type T_P , or in other words, has a certain empirical distribution P . For that, we will use simple combinatorics and some probabilistic arguments. We start from an easy but important fact.

Lemma 1. *For all $x^n \in T_P$, the probability of the sequence under the distribution Q^n , $Q^n(x^n)$, is all the same.*

This is trivial since $Q^n(\cdot)$ depends only on $\hat{P}_{x^n}$. In other words, $Q^n(\cdot)$ is invariant to any permutation of x^n .

Lemma 2 (The number of types).

$$|\mathcal{P}_n| \leq (n+1)^{|\mathcal{X}|}. \quad (5)$$

Remarks:

- 1) The proof of this lemma is trivial. Each distribution $P \in \mathcal{P}_n$ has $|\mathcal{X}|$ entries. Each entry can take value from $\{0, 1/n, 2/n, \dots, 1\}$.
- 2) Obviously, the bound is quite loose, since it ignores the constraint that the sum of entries of P should be 1 to make P a valid distribution. For example, when $\mathcal{X} = \{0, 1\}$, we know that $|\mathcal{P}_n| = n+1$, but this bound gives $(n+1)^2$.
- 3) It turns out that this bound is quite sufficient when we only care about exponential rates, since it increases polynomially in n .
- 4) Recall that these types are disjoint and they cover the entire space $\mathcal{X}^n$. Thus, at least some types must contain exponentially many sequences of $x^n \in \mathcal{X}^n$. [PICTURE]

Theorem 1 (Size of each type). *For convenience, let $|\mathcal{X}| = M$ and let $\mathcal{X} = \{a_1, a_2, \dots, a_M\}$.*

$$\begin{aligned} |T_P| &= \binom{n}{nP(a_1), nP(a_2), \dots, nP(a_M)} \\ &\doteq e^{nH(P)} \end{aligned} \quad (6)$$

To prove this theorem, we will use Stirling's approximation.

Lemma 3 (Stirling's approximation).

$$n! \approx \sqrt{2\pi n} \cdot n^n e^{-n}. \quad (7)$$

More precisely,

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi n} \cdot n^n e^{-n}} = 1. \quad (8)$$

Or even more precisely,

$$\sqrt{2\pi n} \cdot n^n e^{-n} e^{\frac{1}{12n+1}} \leq n! \leq \sqrt{2\pi n} \cdot n^n e^{-n} e^{\frac{1}{12n}}. \quad (9)$$

The proof can be found in Google. We will not need the full strength of this approximation in proving the exponentially increasing rate of the size of each type.

Proof of Theorem 1.

$$\begin{aligned}
|T_P| &= \binom{n}{nP(a_1), nP(a_2), \dots, nP(a_M)} \\
&= \frac{n!}{\prod_{i=1}^M (nP(a_i))!} \\
&\doteq \frac{n^n e^{-n} \sqrt{2\pi n}}{\prod_{i=1}^M (nP(a_i))^{nP(a_i)} e^{-nP(a_i)} \sqrt{2\pi n P(a_i)}} \\
&\doteq \frac{1}{\prod_{i=1}^M P(a_i)^{nP(a_i)}} = \frac{1}{\prod_{i=1}^M e^{nP(a_i) \log P(a_i)}} = e^{nH(P)}.
\end{aligned} \tag{10}$$

□

Remarks:

- 1) One thing we have to be careful when we apply Stirling's approximation is that the number of multiplicative factors should not exceed any constant scaling of n . In proving the above theorem, we have introduced M multiplicative factors and ignored the polynomial term in each factor. However, if we had introduced n factors, for example, the exponential approximation would not have been valid.
- 2) The theorem gives yet another operational meaning of entropy: the exponential size of the corresponding type. So, the entire space $\mathcal{X}^n$ is divided into polynomially many types, and the most of these types have exponential sizes, but with different exponents.

Now we strengthen Lemma 1 to the following. While we know that all sequences in a type have the same probability, evaluated by any product distribution, this lemma gives the precise expression of the probability of each sequence.

Lemma 4. *For any $x^n \in T_P$,*

$$Q^n(x^n) = e^{-n[H(P) + D(P\|Q)]} \tag{11}$$

Proof.

$$\begin{aligned}
Q^n(x^n) &= \prod_{i=1}^n Q(x_i) \\
&= \prod_{a \in \mathcal{X}} Q(a)^{N(a|x^n)} \\
&= \prod_{a \in \mathcal{X}} Q(a)^{nP(a)} \\
&= e^{n \sum_{a \in \mathcal{X}} P(a) [\log Q(a)/P(a) + \log P(a)]} = e^{-n[H(P) + D(P\|Q)]}.
\end{aligned} \tag{12}$$

□

For a special case of $P = Q$, $P^n(x^n) = e^{-nH(P)}$ for $x^n \in T_P$.

Combining (6) and (11), we have the following corollary.

Corollary 1.

$$Q^n(T_P) \doteq e^{-nD(P\|Q)} \tag{13}$$

Remarks:

- 1) This is the most important result of this lecture

- 2) Recall we mentioned that $D(P\|Q)$ is in some sense the “distance” between two distributions P and Q . Now this intuition is made precise. An i.i.d. length- n sequence from an ensemble distribution Q , with n large, would typically have an empirical distribution that is close to Q . Corollary 1 says that the probability that we observe a sequence having an empirical distribution $P \neq Q$ is exponentially small, and the exponent is $D(P\|Q)$. The larger $D(P\|Q)$ is, the faster this probability decays with n .
- 3) Notice that $D(P\|Q) \neq D(Q\|P)$, which implies $Q^n(T_P) \neq P^n(T_Q)$.

Lemma 5.

$$Q^n(T_Q) \geq Q^n(T_P), \quad \forall P, Q \in \mathcal{P}_n \quad (14)$$

Proof. Homework. $\square$

This lemma says that when we generate a length- n sequence by the distribution Q^n , T_Q is the most probable type.

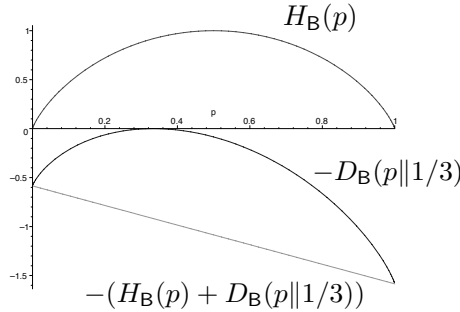


Figure 1: The exponential rates of the size and probability of binary types

Example: [Binary Sequences] Consider length- n binary sequences in $\{0,1\}^n$. Assume that length- n sequences are generated by distribution $Q = \{2/3, 1/3\}$. Consider types for length- n binary sequences and denote the types by $T[p] := T_{\{1-p, p\}}$. The size of $T[p]$ equals $|T[p]| \doteq e^{nH_B(p)}$ where $H_B(p) = -p \log p - (1-p) \log(1-p)$. We plot the exponentially increasing rate of the size of $T[p]$, i.e., $\frac{1}{n} \log |T[p]| = H_B(p)$. The exponent of $T[p]$ is maximized for $p = 1/2$. Next consider the probability of a sequence x^n in type $T[p]$. The exponent of the probability equals $-(H_B(p) + D_B(p\|1/3))$ where $D_B(p\|1/3)$ is the divergence between Bernoulli(p) and Bernoulli($1/3$), i.e., $D_B(p\|1/3) = p \log(p/(1/3)) + (1-p) \log((1-p)/(2/3))$. The most probable sequence in $\{0,1\}^n$ under Q^n is a sequence with all 0. But there exists only one sequence in type $T[1]$. Therefore, when we consider a trade-off between the number of sequences and the probability of sequence, we can find the most probable type. The probability of each type is sum of the two exponents $H_B(p)$ and $-(H_B(p) + D_B(p\|1/3))$, which equals $-D_B(p\|1/3)$. For this example, such a type is $T[1/3]$.

Discussion: For a certain distribution P on $\mathcal{X}$, P might not be in a type $\mathcal{P}_n$. But for large enough n , an arbitrary P on $\mathcal{X}$ becomes a valid type. Moreover, for any $P \in \mathcal{P}_n$, we can always find a $P' \in \mathcal{P}_n$ that is very close to P . Since entropy is a continuous function in the distribution, we can say that $H(P) \approx H(P')$ where P and P' are close. To be more precise, as an example, let's consider a binary case and the type $\{\frac{1}{2} + \frac{1}{n}, \frac{1}{2} - \frac{1}{n}\}$. The size of this type is

$$|T_{\{1/2+1/n, 1/2-1/n\}}| \doteq e^{nH_B(1/2+1/n)}. \quad (15)$$

Moreover, as $n \rightarrow \infty$,

$$|T_{\{1/2+1/n, 1/2-1/n\}}| \doteq e^{nH_B(1/2+1/n)} \doteq |T_{\{1/2, 1/2\}}|. \quad (16)$$

Therefore, we do not quite distinguish between polynomially many distributions (types) that are close enough in the sense that the exponentially increasing rate of the types are the same.

3.2 Method of Types

We next consider the size of union of more than one type. Consider T_P and T_Q with $H(P) > H(Q)$. Remind that types are disjoint.

$$e^{nH(P)} \leq |T_P \cup T_Q| = |T_P| + |T_Q| \doteq e^{nH(P)} + e^{nH(Q)} \leq 2e^{nH(P)} \doteq e^{nH(P)} \quad (17)$$

Therefore, the union has exponential rate $H(P)$, and the sum is dominated by $|T_P|$.

For $P_1, P_2, \dots, P_M$, without loss of generality, assume that $H(P_1) \geq H(P_i) \forall i$. Then,

$$e^{nH(P_1)} \leq \left| \bigcup_{i=1}^M T_{P_i} \right| \leq M|T_{P_1}| \doteq M e^{nH(P_1)}. \quad (18)$$

Therefore, the size of the union of types equals $|\bigcup_{i=1}^M T_{P_i}| \doteq e^{nH(P_1)}$.

Now let $\mathcal{A}$ be a set of distributions. For example, $\mathcal{A} \triangleq \{P : \mathbb{E}_P[F(X)] > \alpha\}$. Consider

$$\left| \bigcup_{P \in \mathcal{A}} T_P \right| = \sum_{P \in \mathcal{A}} |T_P| \doteq \sum_{P \in \mathcal{A}} e^{nH(P)}. \quad (19)$$

Suppose a particular distribution

$$P^* = \arg \max_{P \in \mathcal{A}} H(P). \quad (20)$$

We have

$$e^{nH(P^*)} \leq \left| \bigcup_{P \in \mathcal{A}} T_P \right| \leq (n+1)^{|\mathcal{X}|} e^{nH(P^*)}. \quad (21)$$

Thus,

$$\left| \bigcup_{P \in \mathcal{A}} T_P \right| \doteq e^{nH(P^*)}. \quad (22)$$

Remarks:

- 1) Note that in any set $\mathcal{A}$ there are no more than polynomial-in- n types. This holds only when $|\mathcal{X}|$ is finite. When $\mathcal{X} = \mathbb{R}$ we potentially can have a trouble.
- 2) Exactly the same arguments as above can be used to calculate the probability of union of types, instead of cardinality of union of types.

$$e^{-nD(P^{**} \| Q)} \doteq Q^n(T_{P^{**}}) \leq Q^n \left(\bigcup_{P \in \mathcal{A}} T_P \right) \leq (n+1)^{|\mathcal{X}|} Q^n(T_{P^{**}}) \doteq e^{-nD(P^{**} \| Q)} \quad (23)$$

where

$$P^{**} = \arg \min_{P \in \mathcal{A}} D(P \| Q). \quad (24)$$

[PICTURE: Projection from $\mathcal{Q}$ to set $\mathcal{A}$]

- 3) Note that the cardinality of the union of types is dominated by the type that *maximizes* entropy, and the probability of union of types is dominated by the type that *minimizes* divergence.

- 4) This is the power of the method of types. There are just polynomially many types. Thus when we are interested in the exponential rates, we just need to find a particular type that dominates the rate.

Theorem 2 (Sanov's Theorem). *Let Q be a distribution over $\mathcal{X}$. Let $\mathcal{A}$ be a set of distributions over $\mathcal{X}$, and suppose that $Q \notin \mathcal{A}$.*

$$Q^n \left(\{x^n : \hat{P}_{x^n} \in \mathcal{A}\} \right) \doteq e^{-n \min_{P \in \mathcal{A}} D(P \| Q)}. \quad (25)$$

3.3 Strong Typicality

We next define a particular kind of set $\mathcal{A}$. For a given distribution Q , suppose $Q(x) > 0, \forall x$, and consider

$$\mathcal{A} = \{P : |P(x) - Q(x)| \leq \delta, \forall x \in \mathcal{X}\}, \quad (26)$$

which can be viewed as a δ -neighborhood of a distribution Q .

Definition 4 (Strong Typical Set). *For a given distribution Q with $Q(x) > 0, \forall x$, and a constant $\delta > 0$, the strong typical set is defined as*

$$\tilde{T}_Q = \bigcup_{P \in \mathcal{A}} T_P = \bigcup_{\{P : |P(x) - Q(x)| \leq \delta, \forall x\}} T_P \quad (27)$$

Theorem 3 (Strong AEP). *For any $\epsilon, \delta > 0$, there exists a N_0 s.t. $\forall n > N_0$,*

$$Q^n(\tilde{T}_Q) > 1 - \epsilon. \quad (28)$$

There are two proofs.

Proof 1 of Strong AEP. It is enough to show that

$$\lim_{n \rightarrow \infty} Q^n \left(\bigcup_{P \in \mathcal{A}^c} T_P \right) = 0. \quad (29)$$

□

Note that

$$Q^n \left(\bigcup_{P \in \mathcal{A}^c} T_P \right) \doteq e^{-n \min_{P \in \mathcal{A}^c} D(P \| Q)} \quad (30)$$

For $\delta > 0$, $\mathcal{A}^c$ does not include P and thus $\min_{P \in \mathcal{A}^c} D(P \| Q) > \Delta(\delta) > 0$. Even when we set $\delta_n \rightarrow 0$, which makes $\Delta_n = \Delta(\delta_n) \rightarrow 0$, as long as δ_n decreases slowly, say slower than $n^{-1/2}$, which makes Δ_n decrease much slower than $\frac{1}{n}$, still $\min_{P \in \mathcal{A}^c} D(P \| Q)$ is big enough to make $Q^n(\bigcup_{P \in \mathcal{A}^c} T_P)$ goes to 0 as $n \rightarrow \infty$.

Proof 2 of Strong AEP. What we want to show is

$$Q^n(\{x^n : |\hat{P}_{x^n}(a) - Q(a)| > \delta \text{ for some } a \in \mathcal{X}\}) \rightarrow 0. \quad (31)$$

For any $a \in \mathcal{X}$, by using WLLN to $\mathbb{1}_{X_i=a}$ for $i = 1, \dots, n$, we can show that

$$Q^n \left(\{x^n : |\hat{P}_{x^n}(a) - Q(a)| > \delta\} \right) < \frac{\epsilon}{|\mathcal{X}|} \quad (32)$$

for any $\delta, \epsilon > 0$. By using union bound,

$$\begin{aligned}
& Q^n(\{x^n : |\hat{P}_{x^n}(a) - Q(a)| > \delta \text{ for some } a \in \mathcal{X}\}) \\
& \leq \sum_{a \in \mathcal{X}} Q^n(\{x^n : |\hat{P}_{x^n}(a) - Q(a)| > \delta\}) \\
& \leq \epsilon.
\end{aligned} \tag{33}$$

□

Remarks:

- 1) Strong AEP does not require strong law of large numbers. It simply requires repeated uses of WLLN. The reason it is called Strong AEP is because it gives more detailed description of the typical behavior of DMS strings. Strong AEP says that the empirical distribution of a sequence from $\text{DMS}(Q)$ is, with a high probability, close to Q itself. This closeness is in terms of comparing two distributions, $\hat{P}_{x^n}$ and Q , i.e., $|\hat{P}_{x^n}(x) - Q(x)|$ is small for all $x \in \mathcal{X}$. On the other hand, weak AEP says that with a high probability, $\mathbb{E}_{\hat{P}_{x^n}}[-\log Q(x)]$ is close to $\mathbb{E}_Q[-\log Q(x)] = H(Q)$.
- 2) Consider Bernoulli(1/2) distribution. In the sense of weak AEP, all sequences in $\{0, 1\}^n$ are typical since $x^n = 2^{-n}$ for all $x^n \in \{0, 1\}^n$, which is counter-intuitive. In terms of strong AEP, the typical sequences are only those with roughly half 0's and half 1's. This makes more sense according to our intuition.
- 3) In Proof 1, we showed that $Q^n(\bigcup_{P \in \mathcal{A}^c} T_P) \doteq e^{-n\Delta} \rightarrow 0$. It is a terrible idea to use $Q^n(\bigcup_{P \in \mathcal{A}} T_P) \doteq e^{n \cdot 0}$, since this probability can be any number. Thus, this way cannot prove (28).

Now similar to the weak AEP, we also need to know how large the strong typical set is.

Proposition 1.

$$|\tilde{T}_Q| \doteq e^{nH(Q)}. \tag{34}$$

Proof.

$$|\tilde{T}_Q| = \sum_{P \in \mathcal{A}} |T_P| \doteq e^{nH(P^*)} \tag{35}$$

where

$$P^* = \arg \max_{P \in \mathcal{A}} H(P). \tag{36}$$

Here we use the fact that $\mathcal{A}$ is defined as a neighborhood of Q with radius proportional to δ , and that $H(\cdot)$ is a continuous function, hence $H(P^*) \leq H(Q) + \Delta$, where $\Delta \rightarrow 0$ as $\delta \rightarrow 0$. □

As like the weak AEP, by combining the fact that the strong typical set has size of $\doteq e^{nH(Q)}$ and the strong typical set takes almost all the probability, we can prove the source coding theorem by encoding only the sequences in the strong typical set. The rate of such a source coding is $H(Q)$.

3.4 Large deviations: Sanov's theorem

Theorem 4 (Sanov's Theorem). *Let Q be a distribution over $\mathcal{X}$. Let $\mathcal{A}$ be a set of distributions over $\mathcal{X}$, and suppose that $Q \notin \mathcal{A}$.*

$$Q^n(\{x^n : \hat{P}_{x^n} \in \mathcal{A}\}) \doteq e^{-n \min_{P \in \mathcal{A}} D(P||Q)}. \tag{37}$$

We are interested in Sanov's theorem as it has the following operational meanings. Suppose X^n is a length- n sequence from $\text{DMS}(Q)$. The AEP says that with a high probability, the empirical distribution $\hat{P}_{X^n}$ would lie in a neighborhood of Q . However, with an exponentially small probability, rare event occurs and $\hat{P}_{X^n}$ could be else where, say, lies in a set $\mathcal{A}$. Using the method of types, we can show that the probability of this rare event is dominated by a particular sub-event such that $\hat{P}_{X^n} = P^*$, where P^* is the minimizer of $D(P\|Q)$ for $P \in \mathcal{A}$. That is, given the rare event $\{\hat{P}_{X^n} \in \mathcal{A}\}$ occurs, most likely, $\hat{P}_{X^n} = P^*$. Recall that the atypical event we described here corresponds to the case where an encoder would fail, or makes an error. Thus, knowing the typical way the encoder fails would be valuable.

In fact, we are particularly interested in Sanov's Theorem applied to a special case.

3.5 Universal source coding

Now let's go back to the source coding problem. Both Huffman coding and Shannon coding compress an i.i.d. source with a known distribution. What compression can be achieved if the true distribution P is unknown? Is there a universal code of rate R , say, that suffices to describe every i.i.d. source with entropy $H(P) < R$? The surprising answer is yes and we will talk of this type of universal code in this section. We begin with a definition.

Definition 5. An (n, R) block source code of rate R for a source $X_1, X_2, \dots, X_n$ which has an unknown distribution Q consists of a pair of mapping f_n and g_n :

$$\begin{aligned} f_n : \mathcal{X}^n &\rightarrow \{0, 1\}^{nR} \\ g_n : \{0, 1\}^{nR} &\rightarrow \mathcal{X}^n. \end{aligned} \quad (38)$$

The probability of error for the code with respect to the distribution Q is

$$P_e^{(n)} \triangleq Q^n(\{x^n : g_n(f_n(x^n)) \neq x^n\}). \quad (39)$$

We say that an error exponent E is achievable at rate R if there exists a sequence of (n, R) codes with

$$P_e^{(n)} \leq e^{-nE}, \quad (40)$$

which means

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log P_e^{(n)} \leq -E. \quad (41)$$

Definition 6. A rate R block code for a source will be called universal if the functions f_n and g_n do not depend on the distribution Q and if $P_e^{(n)} \rightarrow 0$ as $n \rightarrow \infty$ if $R > H(Q)$.

We now describe one such universal encoding scheme, that is based on the fact that the number of sequences of type P increases exponentially in the length n of the sequence with the entropy $H(P)$ and the fact that there are only a polynomial number of types.

Theorem 5. There exists a sequence of (n, R) universal source codes such that $P_e^{(n)} \rightarrow 0$ for every source Q such that $H(Q) < R$. With this universal source coding, the achievable error exponent is

$$E = \min_{P: H(P) > R} D(P\|Q), \quad (42)$$

i.e.,

$$P_e^{(n)} \leq e^{-n \min_{P: H(P) > R} D(P\|Q)}. \quad (43)$$

Proof. Consider the set of sequences

$$A = \{x^n \in \mathcal{X}^n : H(\hat{P}_{x^n}) \leq R\} \quad (44)$$

Then

$$\begin{aligned} |A| &= \sum_{P \in \mathcal{P}_n : H(P) \leq R} |T_P| \\ &\doteq \sum_{P \in \mathcal{P}_n : H(P) \leq R} e^{nH(P)} \\ &\leq (n+1)^{|\mathcal{X}|} e^{nR} \end{aligned} \quad (45)$$

Therefore, we can enumerate all the sequences in the set A by a code of rate R . We claim an encoding error when we observe a sequence $x^n \notin A$.

We now show that this encoding scheme is universal. Assume that the distribution of $X_1, X_2, \dots, X_n$ is Q and $H(Q) < R$. Then the probability of decoding error is given by

$$\begin{aligned} P_e^{(n)} &= 1 - Q^n(A) \\ &= \sum_{P: H(P) > R} Q^n(T_P) \\ &\leq (n+1)^{|\mathcal{X}|} \max_{P: H(P) > R} Q^n(T_P) \\ &\leq (n+1)^{|\mathcal{X}|} e^{-n \min_{P: H(P) > R} D(P||Q)}. \end{aligned} \quad (46)$$

□

Remarks:

- 1) One thing we don't like about Huffman code, which is the optimal source code with variable codeword lengths, is that the scheme requires knowledge of the distribution of the DMS. In reality, we often have to compress some source without any statistical knowledge of the source. Thus, a blind, or universal, algorithm is desired. The above scheme is somewhat blind, as it doesn't require any knowledge of the source. Given the rate R , the source encoder enumerates all the sequences whose empirical distribution $\hat{P}_{x^n}$ satisfies $H(\hat{P}_{x^n}) < R$.
- 2) The above universal source coding works well for a DMS. But in reality, we want to have a universal source code that works well even if the source is not a DMS, but with a more complicated statistical structure, such as a Markov process, or even has no statistical structure at all. Surprisingly, this requirement can be achieved by the Lempel-Ziv code. We will not cover it in this course, but you can simply google it or wait until you take a more advanced course about information theory/coding.